Adriane Lau | adriane.lau.e@gmail.com | (760) 704-9893 | Portfolio

EDUCATION

University of California San Diego (UCSD) 2023 - March 2027 (expected)
B.S. Mechanical Engineering, Minor in Cognitive Science Design Cumulative GPA: 3.37

WORK EXPERIENCE

Aynak, Inc. January 2025 - Present
Mechanical Engineering Intern

- Spearheaded design and hardware integration of 0.11 mm thin speaker **flex PCBs**, optimizing **stack-up** thickness and routing through 3 mm heat-set hinges within smart-eyewear frames housing 4 development boards and 1 battery.
- Designed and **SLA** printed custom mounts for 3 distinct speaker configurations, used in experimental testing of beamformer sound transmission. Optimized mount geometries and hardware placement to minimize howling and acoustic feedback.
- Prototyped hearing aid receiver mount integrated into smart eyewear frames of sub-3 mm internal clearance.
- Performed quality assurance and testing of actuators for bone conduction hearing within rigid components of eyewear frame.
- Collaborating with electrical engineer to improve PCB routing and scalable **manufacturability** of Aynak's smart glasses.

Luan Lab, University of California San Diego January 2025 - Present
Undergraduate Researcher https://sites.google.com/ucsd.edu/luangroup

- Implementing mechanical guided assembly into wearable and flexible electronics such as skin sensors and biofeedback mechanisms. Heading experimental procedures on morphing systems and programmable structures.
- Developed Gallium-Indium synthesis approach for electromagnetically reconfigured channels within substrates.
- Fabricated 5 mm wide substrates via **photolithography** and 4 custom **PLA** mold patterns designed in **SolidWorks**.
- Aim to develop novel electromagnetism-based methods for patient specific morphing healthcare devices.

MAE 3 Course Tutor March 2025 - Present
Design Studio Tutor

- Constructed "World cup" robot combat arena in **AutoCAD**, **lasercut** and assembled with T-Slot frames and material constraints.
- Redesigned 1 of 2 project fabrication procedures for **Fusion 360 CAD** of pendulum clock, developing smoothed replacement **CAD** models, creating labeled graphics and video tutorials, and modeling common errors with prevention strategies.
- Designed a geared motor adapter and demo for ongoing lab curriculum to demonstrate press fit and linkages in lecture.
- Maintain fabrication studio, assisted in instrumentation troubleshooting, and maintain **LaserCAMM** and Bambu **3D printers**.

Instructional Assistant, Lab Tutor

- Provided mentorship to a total of 19 students on mechanical design and **GD&T**, leading 2 fabrication projects in studio.
- Taught manufacturing, fabrication, and **CAD** skills, including **Fusion 360**, **AutoCAD**, laser cutting, and **3D printing**. Developed lab agendas and trainings to guide students in brainstorming, designing, and building competitive robots.

Vi Senior Living Server and Host July 2023 - April 2026
Server. Adapted to dual roles as server and hostess

RELEVANT PROJECTS

Portable Volleyball Flywheel Summer 2025
Personal Project

- Conceptualized a **SolidWorks CAD** model of portable volleyball serving/tossing flywheel for personal use, collaborating with Voyager aerospace engineer on motor specs, industry standards, and fabrication critiques. Developed **MATLAB** script to determine linear travel distance and trajectory of ball, **technical drawings**, and **motion analysis** with **GD&T standards**.

Triton Robotics January 2024 - June 2025
UCSD Robotics Team, Infantry Mechanical Division | tritonrobotics.org

- Supported Triton Robotics' first time hosting RMNA competition, manufacturing playing fields and obstacles for 28 universities.
- Redesigned armor mount with interference fit for **ease of assembly** and time efficiency during competition. Integrated a **3D printed (PLA)** snap-fit joint to mount and accommodate new slip ring using **OnShape**.
- Developed a **SolidWorks** model Omni wheel into a new suspension system, optimizing it for a new motor assembly of chassis.
- Initiation project: designed bearing armor cover with rotational functionality to serve as protection during competition

SKILLS

Technical Skills: CAD, GD&T, DFM, FEA, tolerance analysis, mechanical prototyping, machine shop, Python (Pandas), MATLAB
Software & Tools: SolidWorks (CAM, motion analysis, FEA), Fusion 360 (Mesh, Surface, Sheet Metal), Onshape, AutoCAD, ANSYS (FEA), MATLAB, Jupyter Notebooks, Excel, Visual Studio Code
Fabrication: SLA/FDM 3D printing, laser cutting, heat-set inserts, flex PCB integration